



Principles of Finance

Professor Robert Kosowski

MSc Climate Change, Management and Finance

Week 5

Module Overview

Week 1

Introduction, Institutions, Financial Decision making, Time value of money, Net Present Value (NPV) Rule, Cost-Benefit Analysis (**BD 1-3**)*

Week 2

Time-Value of Money (Perpetuities/Annuities), Interest Rates, Valuing Bonds (**BD 4-6**)*

Week 3

Investment Decision Rules, Fundamentals of Capital Budgeting, Valuing Stocks (**BD 7.1, 7.2 (first part), 8, 9**)*, **Saito Solar case**

Deadline for submission of group assignment 1 is 9am, Thursday, 13th November

Week 4

Capital Markets and Pricing of Risk (**BD 10**), Optimal Portfolio Choice and the CAPM (**BD 11 excl. Appendix**), Estimating the Cost of Capital (**BD 12.1-13.3, excl. Appendix**), Robeco (carbon beta example)

Week 5

Capital Structure, Debt and Taxes (**BD 12.4, 12.5, 12.6, 12.7, BD 14.1, 14.2, 14.3**), Investor Behaviour and Capital Market Efficiency (**BD 13.1, 13.2, 13.6-13.8 excl. Appendix**), Introduction to the balance sheet and asset-liability-management (ALM) of a bank, insurance company and pension fund, **M&M Pizza Case**

*(See **reading list** for detailed chapter references and what is excluded as well as additional optional readings)

Deadline for submission of group assignment 2 is 9am, Tuesday, 2nd December

Motivating examples



Duke Explores Private Credit in Potential First for Utilities



Duke Energy Corp. trucks in Charlotte, North Carolina. Photographer: Callaghan O'Hare/Bloomberg



Week 5 Lectures

Learning objectives

List of topics

Appendix:

Differences between balance sheets and risks of insurers and banks

BD Chapter 12

- Estimate the cost of debt

BD Chapter 13

- How to calculate alpha and understand the effect of competition in capital markets
- Understanding how to test the CAPM
- Style-Based Techniques and Multi-factor models of risk

BD Chapter 14

- Describe the capital structure that the firm should choose.
- List the three conditions that make capital markets perfect.
- Discuss the implications of MM Proposition I, and the roles of homemade leverage and the Law of One Price in the development of the proposition.
- Calculate the cost of capital for levered equity according to MM Proposition II.
- Simple tax shield and (after-tax) WACC calculation

Appendix:

- Differences between balance sheets and risks of insurers and banks

Chapter 12

Learning Objectives

- Estimate the cost of debt, given a company's yield to maturity, probability of default, and expected loss rate.
- Discuss the difference between the yield to maturity and the cost of debt when there is low versus high default risk.
- Calculate the cost of debt given a company's debt beta, the risk free rate, and the market risk premium.
- Illustrate the use of comparable companies' unlevered betas or unlevered cost of capital to estimate project cost of capital.
- Discuss the advantages of using several companies' betas to estimate a project beta.
- Calculate the weighted average cost of capital.
- Discuss strengths and weaknesses of the CAPM.

12.4 The Debt Cost of Capital

Table 12.3 Average Debt Betas by Rating and Maturity*

- Debt Yields Versus Returns
 - Yield to maturity is the IRR an investor will earn from holding the bond to maturity and receiving its promised payments.
 - If there is little risk the firm will default, yield to maturity is a reasonable estimate of investors' expected rate of return.
 - If there is significant risk of default, yield to maturity will overstate investors' expected return.
 - The following table has some real-world estimates of debt beta for more realistic approaches

By Rating	A and above	BBB	BB	B	CCC
Avg. Beta	<0.05	0.10	0.17	0.26	0.31
By Maturity	(BBB and above)	1-5 year	5-10 year	10-15 year	>15 year
Avg. Beta		0.01	0.06	0.07	0.14

12.5 A Project's Cost of Capital

- All-Equity Comparables
 - Find an all-equity financed firm in a single line of business that is comparable to the project.
 - Use the comparable firm's equity beta and cost of capital as estimates.
- Levered Firms as Comparables
 - For levered firms, the cash flows generated by the firm's assets are used to pay both debt and equity holders.
 - The returns of the firm's equity alone are not representative of the underlying assets; Because of the firm's leverage, the equity will often be much riskier.
 - The beta of a levered firm's equity will not be a good estimate of the beta of its assets and of our project.

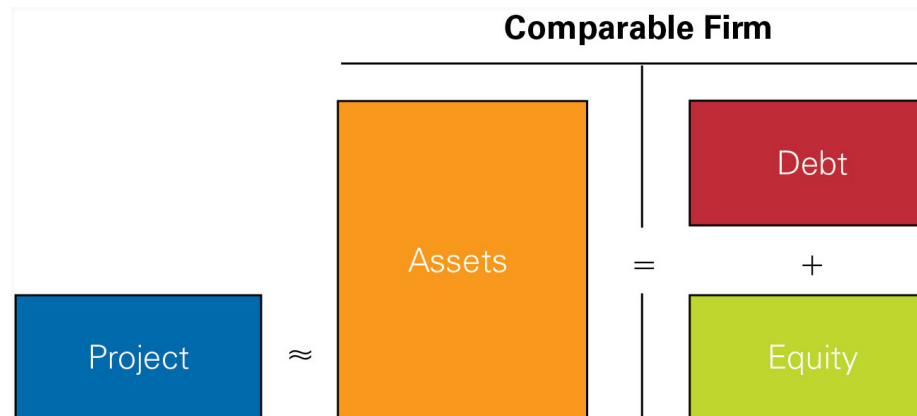


Figure 12.3 Using a Levered Firm as a Comparable for a Project's Risk

12.5 A Project's Cost of Capital

- As we saw in Chapter 11, the expected return of a portfolio is equal to the weighted average of the expected returns of the securities in the portfolio,

The Unlevered Cost of Capital

- Expected return required by investors to hold the firm's underlying assets
- The weighted average of the firm's equity and debt costs of capital can be calculated as:

- Asset or Unlevered Cost of Capital:
$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D$$

Unlevered Beta

- Because the beta of a portfolio is the weighted-average of the betas of the securities in the portfolio, we have a similar expression for the firm's asset or unlevered beta, which we can use to estimate the beta of our project:

- Asset or Unlevered Beta:
$$\beta_U = \frac{E}{E + D} \beta_E + \frac{D}{E + D} \beta_D$$

Estimate an Industry Asset Beta

Textbook Example 12.7

Problem

Consider the following data for U.S. department stores in mid-2009, showing the equity beta, ratio of net debt to enterprise value (D/V), and debt rating for each firm.

Estimate the average and median asset beta for the Dillard's

Company	Ticker	Equity Beta	D/V	Debt Rating
Dillard's	DDS	2.38	0.59	B
JCPenney	JCP	1.60	0.17	BB
Kohl's	KSS	1.37	0.08	BBB
Macy's	M	2.16	0.62	BB
Nordstrom	JWN	1.94	0.35	BBB
Saks	SKS	1.85	0.50	CCC
Sears Holding	SHLD	1.36	0.23	BB

By Rating	A and above	BBB	BB	B	CCC
Avg. Beta	<0.05	0.10	0.17	0.26	0.31
By Maturity	(BBB and above)	1-5 year	5-10 year	10-15 year	>15 year
Avg. Beta		0.01	0.06	0.07	0.14

Solution

Using the data for debt betas from Table 12.3, we can apply Eq. 12.9 for each firm.

$$\beta_U =$$

Estimate an Industry Asset Beta

Textbook Example 12.7

Solution:

Doing this calculation for each firm, we obtain the following estimates:

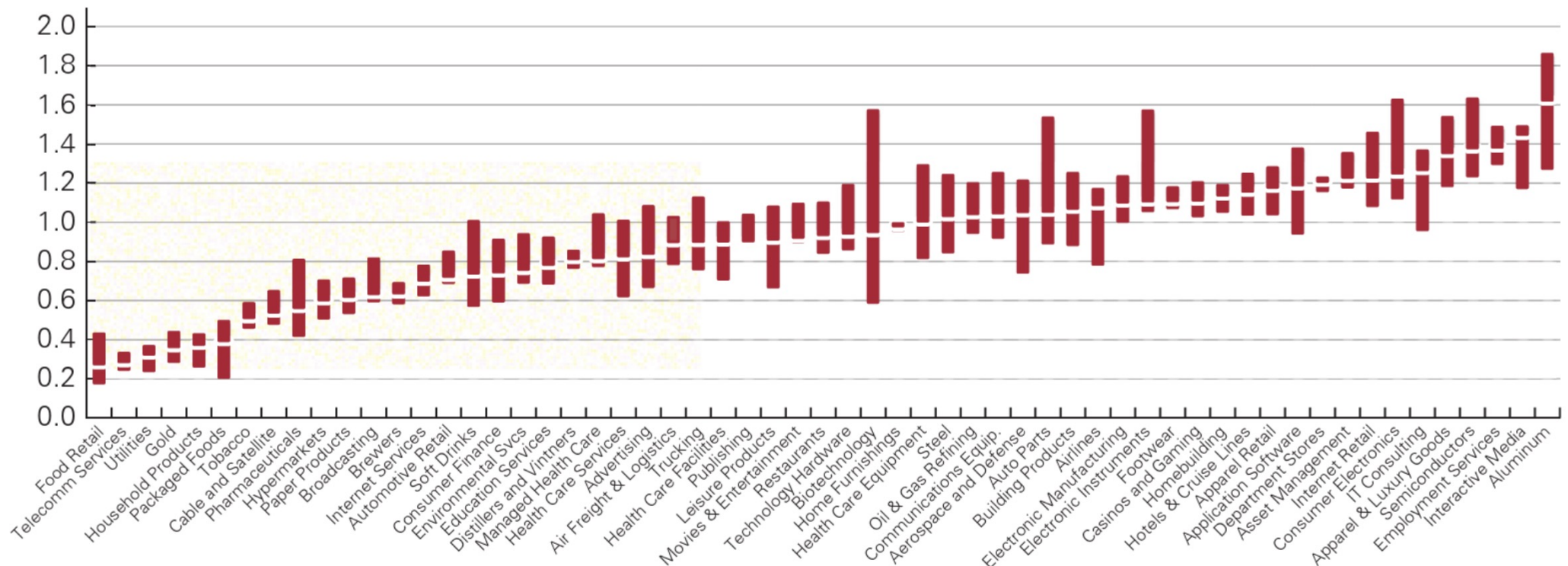
Ticker	Equity Beta	D/V	Debt Rating	Debt Beta	Asset Beta
DDS	2.38	0.59	B	0.26	1.13
JCP	1.60	0.17	BB	0.17	1.36
KSS	1.37	0.08	BBB	0.10	1.27
M	2.16	0.62	BB	0.17	0.93
JWN	1.94	0.35	BBB	0.10	1.30
SKS	1.85	0.50	CCC	0.31	1.08
SHLD	1.36	0.23	BB	0.17	1.09
				Average	1.16
				Median	1.13

The large difference in the firms' equity beta are mainly due to differences in leverage. The firms' asset beta are much more similar, suggesting that the underlying business in this industry have similar market risk. By combining estimates from several closely related firms in this way can get a more accurate estimate of the beta for investments in the industry.

Figure 12.4 Industry Asset Betas (2022)

The plot shows the median, as well as upper and lower quartiles, for estimated asset betas of S&P 1500 firms in selected industries. Note the low asset betas for less cyclical industries such as utilities and household products, versus the much higher asset betas of technology firms, asset management, and capital-intensive cyclical industries (like homebuilding, semiconductors and aluminum).

Asset Beta

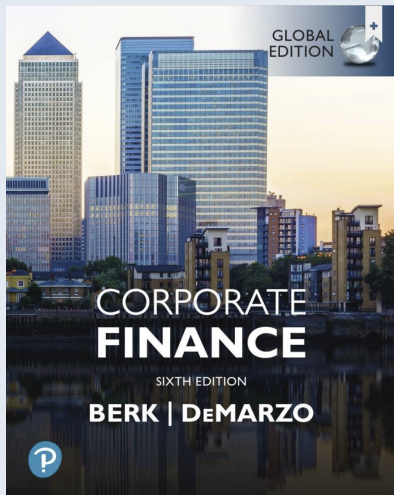


Asset or Unlevered Beta:
$$\beta_U = \frac{E}{E + D} \beta_E + \frac{D}{E + D} \beta_D$$

Chapter 12 Quiz

1. Why does the yield to maturity of a firm's debt generally overestimate its debt cost of capital?
2. What data can we use to estimate the beta of a project?
3. Why does the equity beta of a levered firm differ from the beta of its assets?
4. Why might projects within the same firm have different costs of capital?
5. Even if the CAPM is not perfect, why might we continue to use it in corporate finance?

Chapter 13 (BD)



Chapter Outline

13.1 Competition and Capital Markets

13.2 Information and Rational Expectations

13.6 Style-Based Techniques and the Market Efficiency Debate

13.7 Multifactor Models of Risk

13.8 Methods Used in Practice

Chapter 13

Learning Objectives

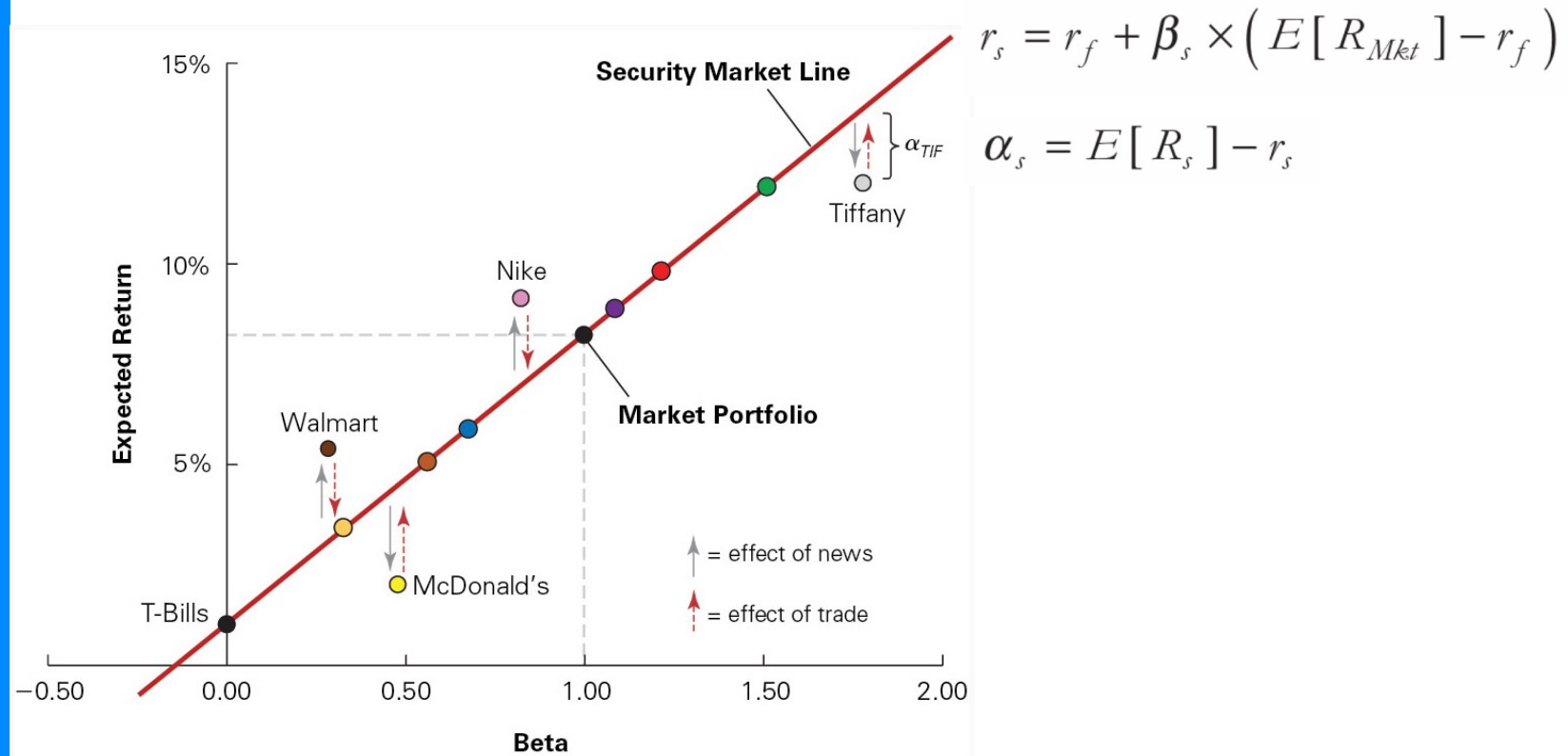
- Compute a stock's alpha.
- Explain how investors' attempts to "beat the market" should keep the market portfolio efficient.
- Describe the effect of homogeneous expectations on a security's alpha.
- Explain why holding the market portfolio does not depend on the quality of an investor's information or trading skills.
- Understand what the CAPM requires about investors' expectations.
- Discuss the size effect.
- Describe the momentum trading strategy.
- Describe the Arbitrage Pricing Theory.
- Discuss the expected return on a self-financing portfolio.
- Discuss the Fama-French-Carhart model.

13.1 Competition and Capital Markets

Figure 13.2 Deviations from the Security Market Line

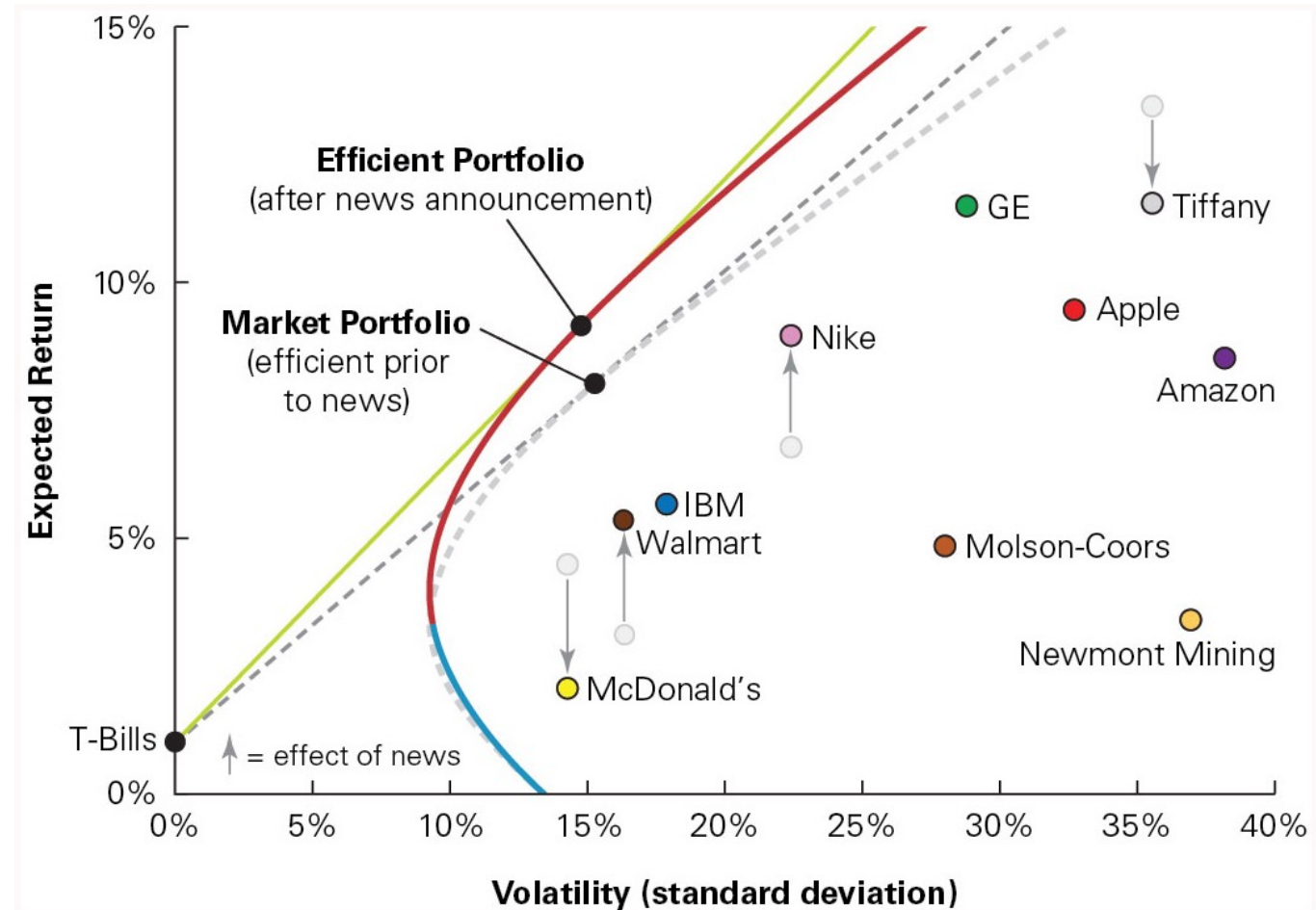
To understand the role of competition in the market, it is useful to consider how the CAPM equilibrium we derived in Chapter 11 might arise based on the behavior of individual investors. In this section, we explain how investors who care only about expected return and variance react to new information and how their actions lead to the CAPM equilibrium.

- Profiting from Non-Zero Alpha Stocks
 - Investors can improve the performance of their portfolios by buying stocks with positive alphas and by selling stocks with negative alphas.
 - Recall that the difference between a stock's expected return and its required return according to the security market line is the stock's alpha:



13.1 Competition and Capital Markets

The arrival of news about the stocks will also shift the expected returns of individual stocks and the market

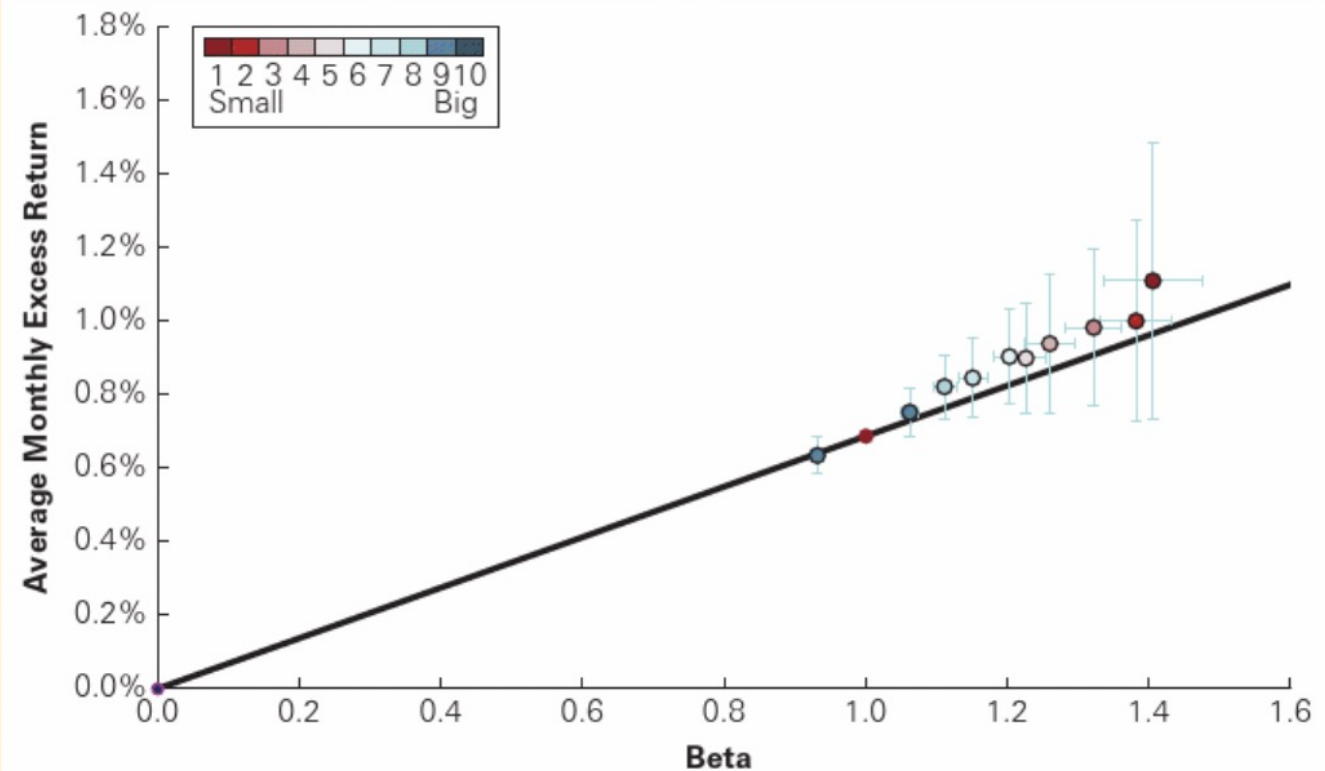


13.6 Style-Based Techniques and the Market Efficiency Debate

The plot shows the average monthly excess return (the return minus the one-month riskfree rate) for 10 portfolios formed in each year based on firms' market capitalizations, plotted as a function of the portfolio's estimated beta. The black line is the security market line. If the market portfolio is efficient and there is no measurement error, all portfolios would plot along this line. The error bars mark the 95% confidence bands of the beta and expected excess return estimates.

As we saw earlier the CAPM can be tested by studying the relationship between average excess returns on an asset or portfolio and beta. There is some evidence that variables (such as M, B/M,...) other than market beta are related to average turns.

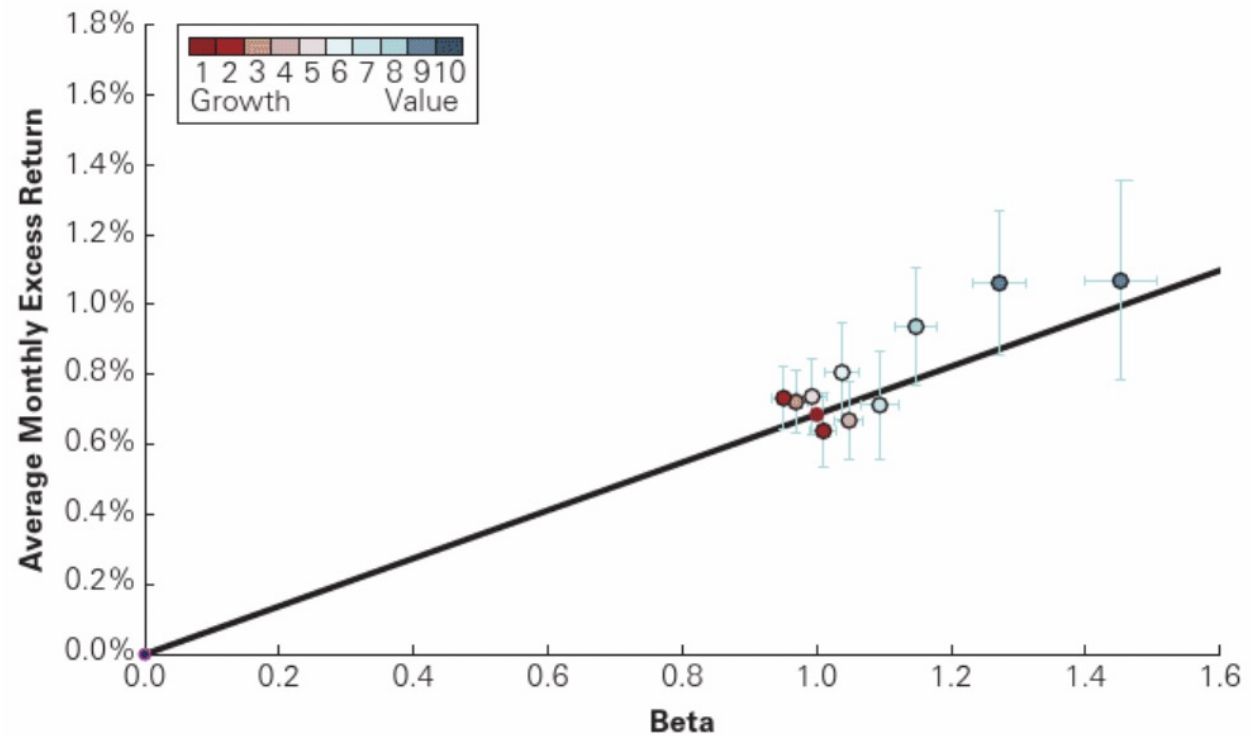
FIGURE 13.9 Excess Return of Size Portfolios, 1926–2022



13.6 Style-Based Techniques and the Market Efficiency Debate

Figure 13.10

Excess Return of Book-to-Market Portfolios, 1926–2022
The plot shows the same data as Figure 13.9, with portfolios formed based on stocks' book-to-market ratios rather than size. Note the tendency of value stocks (high book-to-market) to be above the security market line, and growth stocks (low book-to-market) to be near or below the line.



There is an active debate as to why variables other than CAPM beta explain average excess return differences cross-sectionally:

- (1) a rational expectations explanations states that the variables capture risk and markets are efficient
- (2) A behavioural explanation states that the variables capture behavioural biases
- (3) Other explanations are statistical and related to the difficulty of testing the CAPM

13.7 Multifactor Models of Risk

- Using Factor Portfolios
 - Single-Factor Model
 - A model that uses one portfolio
 - Multi-Factor Model
 - A model that uses more than one portfolio in the model
 - The CAPM is an example of a single-factor model while the **Arbitrage Pricing Theory (APT)** is an example of a multifactor model.

- Using Factor Portfolios
 - Given N factor portfolios with returns $R_{F1}, \dots, R_{FN}$, the expected return of asset s is defined as follows:

$$\begin{aligned} E[R_s] &= r_f + \beta_s^{F1}(E[R_{F1}] - r_f) + \beta_s^{F2}(E[R_{F2}] - r_f) + \dots + \beta_s^{FN}(E[R_{FN}] - r_f) \\ &= r_f + \sum_{n=1}^N \beta_s^{Fn}(E[R_{Fn}] - r_f) \end{aligned}$$

- $\beta^1 \dots \beta^N$ are the factor betas

13.7 Multifactor Models of Risk

Table 13.1 FFC Portfolio Average Monthly Returns, 1927–2018

SMB is a Market Capitalization Strategy, a portfolio of regularly rebalanced small minus big firms

HML is a Book-to-Market Ratio Strategy, a portfolio of regularly rebalanced value minus growth firms

PR1YR is a momentum strategy, a portfolio of regularly rebalanced winner minus loser stocks

- Selecting the Portfolios
 - Fama-French-Carhart (FFC) Factor Specifications

$$E[R_s] = r_f + \beta_s^{Mkt}(E[R_{Mkt}] - r_f) + \beta_s^{SMB}E[R_{SMB}] + \beta_s^{HML}E[R_{HML}] + \beta_s^{PR1YR}E[R_{PR1YR}]$$

Factor Portfolio	Average Monthly Return (%)	95% Confidence Band (%)
Mkt – r_f	0.66	±0.32
SMB	0.26	±0.19
HML	0.38	±0.21
PR1YR	0.66	±0.28

Source: Data courtesy of Kenneth French
mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Although it is widely used in research to measure risk, there is much debate about whether the FFC factor specification is really a significant improvement over the CAPM.

Using the FFC Factor Specification to Calculate the Cost of Capital

Textbook Example 13.3

Problem

You are considering making an investment in a project in the fast food industry. You determine that the project has the same level of non-diversifiable risk as investing in McDonald's stock. **Determine the cost of capital by using the FFC factor specification.**

You decide to use data over the past ten years to **estimate the factor betas** of McDonald's stock (ticker: MCD). Therefore, you regress the monthly excess return (the realized return in cash month minus the risk-free rate) of McDonald's stock on the return of each portfolio. The coefficient estimates are the factor betas. Here are the estimates of the four factor betas and their 95% confidence interval based on data from 2007–2017:

Factor	Beta Estimate	Lower 95%	Upper 95%
Mkt	0.59	0.44	0.75
SMB	−0.53	−0.81	−0.25
HML	−0.02	−0.27	0.24
PR1YR	0.13	−0.01	0.27

Assume a risk-free monthly rate of $2.4\%/12 = 0.20$.

What is cost of capital (expressed as an Annual Percentage Rate (APR)) for McDonald's by using the FFC factor specification?

Using the FFC Factor Specification to Calculate the Cost of Capital

Textbook Example 13.3

Solution

Using the beta estimates and a risk-free monthly rate of $2.4\%/12 = 0.20\%$, you can calculate the monthly expected return for investing in McDonalds's stock:

$$E[R_{MCD}] =$$

=

=

Expressed as an APR, the expected return is

Thus, the annual cost of capital of the investment opportunity is about

As a comparison, a **standard CAPM regression** over the same time period leads to an estimated market beta of 0.45 for McDonalds's—the market beta differs from the estimate of above because we are using only a single factor in the CAPM regression. Using the historical excess return on the market implies an expected return on the market implies an expected return of $0.20\% + 0.45 \times 0.66\% = 0.50\%$ per month, or about 6.0% per year.

Figure 13.2. Which Risk Model Determines Mutual Fund Inflows

This figure reports the frequency that the sign of fund inflows match the sign of last quarter's risk-adjusted return based on alternative risk models. If fund inflows and outflows were random, the frequency would be 50%.

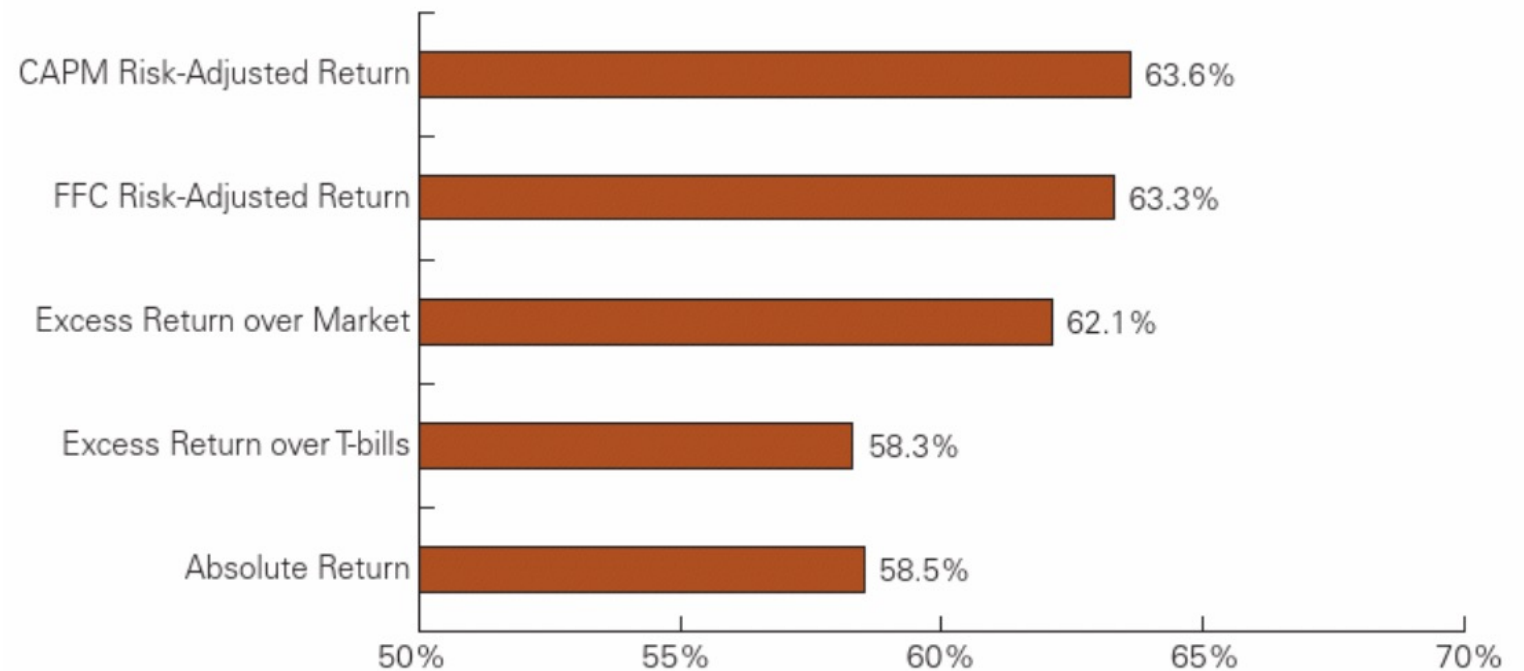
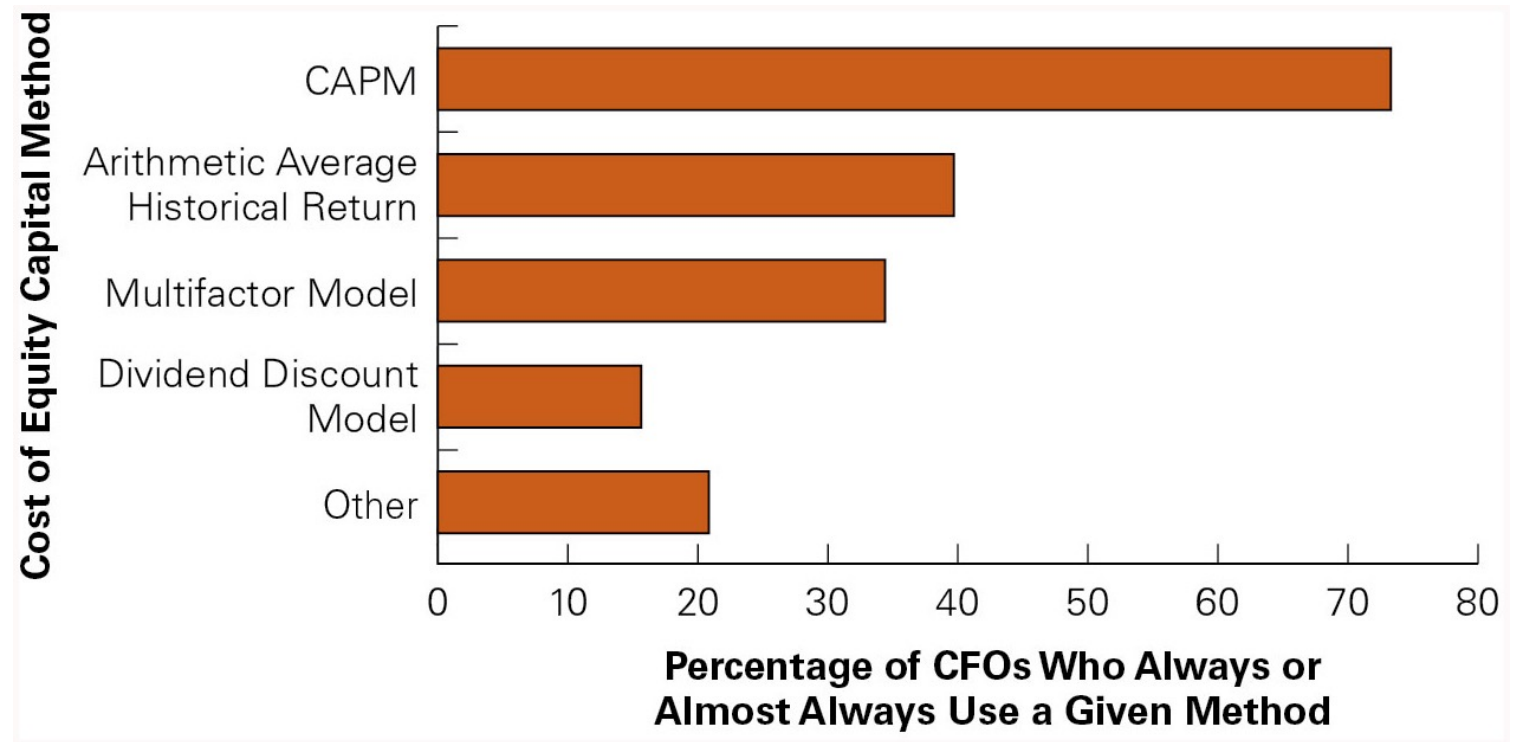


Figure 13.11 How Firms Calculate the Cost of Capital

Note that in a multi-factor model one could include variables such as valuation ratios (B/M) or size or momentum, but also **emissions related variables**.

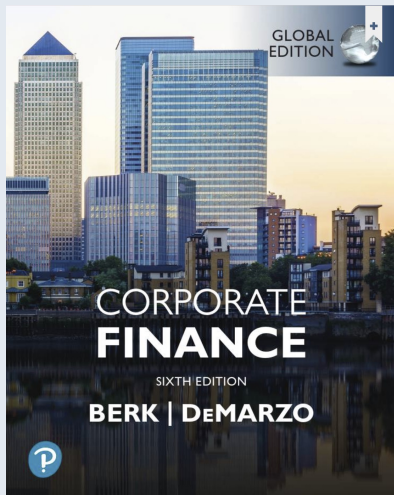
Next term you will study panel regressions that examine whether there is a relationship between emissions and firm's cost of capital controlling for CAPM beta, valuation ratios and other variables

See for example: Bolton and Kacperczyk (2020), "Do Investors Care about Carbon Risk?"



Source: J. R. Graham and C. R. Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," **Journal of Financial Economics** 60 (2001): 187–243.

Chapter 14 (BD)



Chapter Outline

14.2 Modigliani–Miller I: Leverage, Arbitrage, and Firm Value

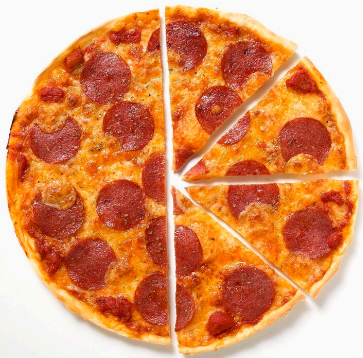
14.3 Modigliani–Miller II: Leverage, Risk, and the Cost of Capital

Chapter 14

Learning Objectives

- List the three conditions that make capital markets perfect.
- Discuss the implications of MM Proposition I, and the roles of homemade leverage and the Law of One Price in the development of the proposition.
- Calculate the cost of capital for levered equity according to MM Proposition II.
- Illustrate the effect of a change in debt on weighted average cost of capital in perfect capital markets.
- Calculate the market risk of a firm's assets using its unlevered beta.
- Illustrate the effect of increased leverage on the beta of a firm's equity.

14.1 Financing a Firm with Debt and Equity



- **Modigliani and Miller** argued that with **perfect capital markets**, the total value of a firm should not depend on its capital structure.
 - Assume the following Date 0 investment and Date 1 cash flows which are **equally likely** as well as a 10% risk premium over the current risk-free interest rate of 5% to invest in this project.

Date 0	Date 1	
	Strong Economy	Weak Economy
-\$800	\$1400	\$900

- The PV and NPV of the cash flows is \$1000 and \$200, respectively, since $PV(\text{equity cash flows}) = \frac{0.5 \times \$1400 + 0.5 \times \$900}{1.15} = \frac{\$1150}{1.15} = \$1000$. If the project was equity financed then $E = \$1000$.
- MM observed that because the cash flows of the debt and equity sum to the cash flows of the project, **in the absence of transaction costs and taxes**, by the Law of One Price the combined values of debt and equity must be \$1000.
 - Therefore, if the value of the debt is \$500, the value of the levered equity must be \$500.
 - $E = \$1000 - \$500 = \$500$
- Because the cash flows of levered equity are smaller than those of unlevered equity, levered equity will sell for a lower price (\$500 versus \$1000).
 - However, you are not worse off.
 - You will still raise a total of \$1000 by issuing both debt and levered equity.
 - Consequently, you would be indifferent between these two choices for the firm's capital structure.

14.1 The Effect of Leverage on Risk and Return

Table 14.4 Returns to Equity with and without Leverage

- Leverage increases the risk of the equity of a firm.
 - Therefore, it is inappropriate to discount the cash flows of levered equity at the same discount rate of 15% that you used for unlevered equity. Investors in levered equity will require a higher expected return to compensate for the increased risk.

	Date 0	Date 1: Cash Flows		Date 1: Returns		
	Initial Value	Strong Economy	Weak Economy	Strong Economy	Weak Economy	Expected Return
Debt	\$500	\$525	\$525	5%	5%	5%
Levered equity	\$500	\$875	\$375	75%	−25%	25%
Unlevered equity	\$1000	\$1400	\$900	40%	−10%	15%

- The returns to equity holders are very different with and without leverage.
 - Unlevered equity has a return of either 40% or 10%, for an expected return of 15%.
 - Levered equity has higher risk, with a return of either 75% or −25%.
 - To compensate for this risk, levered equity holders receive a higher expected return of 25%.
 - Leverage increases the risk of equity even when there is no risk that the firm will default.
 - While debt may be cheaper, its use raises the cost of capital for equity.
 - Considering both sources of capital together, the firm's average cost of capital with leverage is the same as for the unlevered firm.

14.2 The Market Value Balance Sheet

- Market Value Balance Sheet
 - A balance sheet where:
 - **All assets and liabilities** of the firm are included (even intangible assets such as reputation, brand name, or human capital that are missing from a standard accounting balance sheet).
 - All values are current market values rather than historical costs.
 - The total value of all securities issued by the firm must equal the total value of the firm's assets.
- Using the market value balance sheet, the value of equity is computed as follows:

Market Value of Equity =

Market Value of Assets – Market Value of Debt and Other Liabilities

14.2 Application: A Leveraged Recapitalization

- Leveraged Recapitalization
 - When a firm uses borrowed funds to pay a large special dividend or repurchase a significant amount of outstanding shares
- Example:
 - Harrison Industries is currently an all-equity firm operating in a perfect capital market, with **50 million shares outstanding** that are trading for \$4 per share.
 - Harrison plans to increase its leverage by borrowing **\$80 million** and using the funds to repurchase 20 million of its outstanding shares.
- This transaction can be viewed in two stages.
 - First, Harrison sells debt to raise \$80 million in cash.
 - Second, Harrison uses the cash to repurchase shares.
- What will be effect of the leveraged recapitalization on Harrison's share price?

14.2 Application: A Leveraged Recapitalization

The balance sheet before and after

Initial		After Borrowing		After Share Repurchase	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
		Cash	Debt	Cash	Debt
		80	80	0	80
Existing assets	Equity	Existing assets	Equity	Existing assets	Equity
200	200	200	200	200	120
200	200	280	280	200	200
Shares outstanding (million)	50	Shares outstanding (million)	50	Shares outstanding (million)	30
Value per share	\$4.00	Value per share	\$4.00	Value per share	\$4.00

14.2 Application: A Leveraged Recapitalization

Table 14.9 Market Value
Balance Sheet after Each
Stage of Harrison's Leveraged
Recapitalization (\$ millions)

- Solution
 - Initially, Harrison is an all-equity firm and the market value of Harrison's equity is \$200 million (50 million shares $\times$ \$4 per share = \$200 million) and equals the market value of its existing assets.
 - After borrowing, Harrison's liabilities grow by \$80 million, which is also equal to the amount of cash the firm has raised.
 - Because both assets and liabilities increase by the same amount, the market value of the equity remains unchanged.

14.2 Application: A Leveraged Recapitalization

- Example
 - To conduct the share repurchase, Harrison spends \$80 million in borrowed cash to repurchase 20 million shares.

$$-\frac{\$80 \text{ million}}{\$4 \text{ per Share}} = 20 \text{ million shares}$$

- Because the firm's assets decrease by \$80 million and its debt remains unchanged, the market value of the equity must also fall by \$80 million, from \$200 million to \$120 million, for assets and liabilities to remain balanced.
- Example
 - The share price is unchanged
 - With 30 million shares remaining, the shares are worth \$4 per share, just as before.

$$\frac{\$120 \text{ million}}{30 \text{ million Share}} = \$4 \text{ per shares}$$

14.3 Modigliani–Miller II: Leverage, Risk, and the Cost of Capital

- Leverage and the Equity Cost of Capital
 - MM's first proposition can be used to derive an explicit relationship between leverage and the equity cost of capital.
- Leverage and the Equity Cost of Capital
 - E
 - Market value of equity in a levered firm
 - D
 - Market value of debt in a levered firm
 - U
 - Market value of equity in an unlevered firm
 - A
 - Market value of the firm's assets

Leverage and the Equity Cost of Capital

MM Proposition I states that $E + D = U = A$.

The total market value of the firm's securities is equal to the market value of its assets, whether the firm is unlevered or levered.

14.3 14.3 Modigliani–Miller II: Leverage, Risk, and the Cost of Capital

- Leverage and the Equity Cost of Capital
 - The cash flows from holding unlevered equity can be replicated using homemade leverage by holding a portfolio of the firm's equity and debt.
 - The return on unlevered equity (R_U) is related to the returns of levered equity (R_E) and debt (R_D):

$$\frac{E}{E + D} R_E + \frac{D}{E + D} R_D = R_U$$

- Multiply LHS and RHS by (E+D)

– Solving for R_E :

$$R_E = \underbrace{R_U}_{\text{Risk without leverage}} + \underbrace{\frac{D}{E} (R_U - R_D)}_{\text{Additional risk due to leverage}} \quad (\text{Eq. 14.5})$$

- The levered equity return equals the unlevered return, plus a premium due to leverage.

The amount of the premium depends on the amount of leverage, measured by the firm's market value debt–equity ratio $\frac{D}{E}$.

14.3 Modigliani–Miller II: Leverage, Risk, and the Cost of Capital

- Leverage and the Equity Cost of Capital
 - *MM Proposition II*:
 - The cost of capital of levered equity is equal to the cost of capital of unlevered equity plus a premium that is proportional to the market value debt–equity ratio
 - Cost of Capital of Levered Equity

$$r_E = r_U + \frac{D}{E}(r_U - r_D) \quad (\text{BD Equation 14.5})$$

- Leverage and the Equity Cost of Capital
 - Recall from above:
 - If the firm is all–equity financed, the expected return on unlevered equity is 15%.
 - If the firm is financed with \$500 of debt, the expected return of the debt is 5%.
- Leverage and the Equity Cost of Capital
 - Therefore, according to MM Proposition II, the expected return on equity for the levered firm is

$$r_E = 15\% + \frac{500}{500}(15\% - 5\%) = 25\%$$

Capital Budgeting and the Weighted Average Cost of Capital

- If a firm is unlevered, all of the free cash flows generated by its assets are paid out to its equity holders
 - The market value, risk, and cost of capital for the firm's assets and its equity coincide and therefore

$$r_U = r_A$$

- If a firm is levered, project r_A is equal to the firm's weighted average cost of capital.
 - Unlevered Cost of Capital (Pretax WACC)

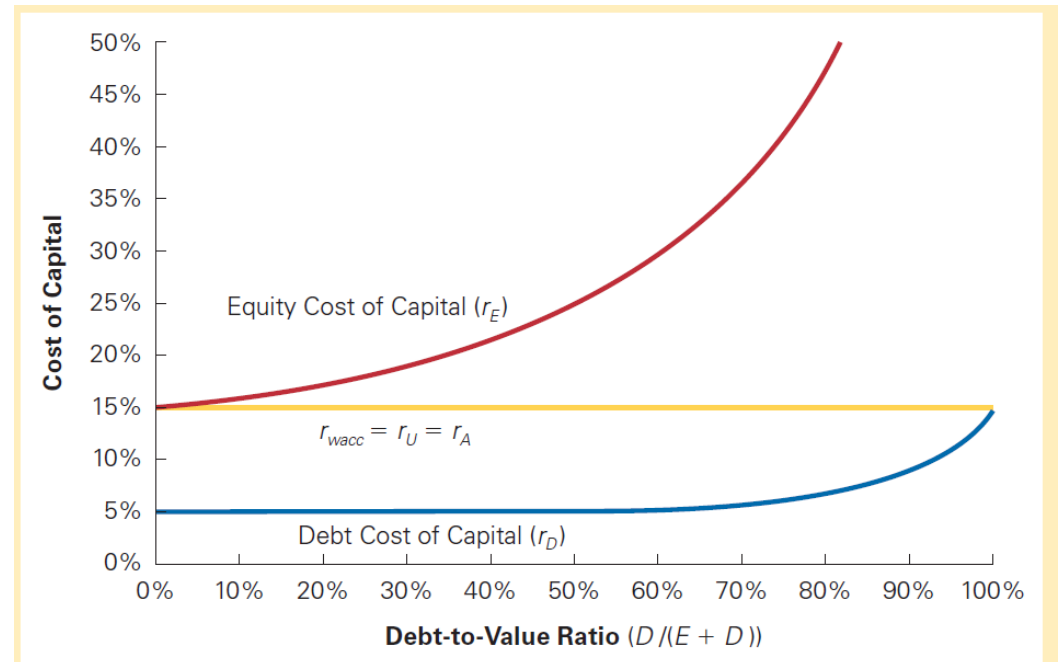
$$\begin{aligned} r_{wacc} &= \left(\frac{\text{Fraction of Firm Value}}{\text{Financed by Equity}} \right) \left(\frac{\text{Equity}}{\text{Cost of Capital}} \right) + \left(\frac{\text{Fraction of Firm Value}}{\text{Financed by Debt}} \right) \left(\frac{\text{Debt}}{\text{Cost of Capital}} \right) \\ &= \frac{E}{E + D} r_E + \frac{D}{E + D} r_D \end{aligned}$$

$$r_{wacc} = r_U = r_A$$

That is, with perfect capital markets, a firm's WACC is independent of its capital structure and is equal to its equity cost of capital if it is unlevered, which matches the cost of capital of its assets .

WACC and Leverage with Perfect Capital Markets

- (a) Equity, debt, and weighted average costs of capital for different amounts of leverage. The rate of increase of r_D and r_E , and thus the shape of the curves, depends on the characteristics of the firm's cash flows.
- (b) Calculating the WACC for alternative capital structures. Data in this table correspond to the example in Section 14.1.



(a)

E	D	r_E	r_D	$\frac{E}{E+D}r_E + \frac{D}{E+D}r_D$	$= r_{WACC}$
1000	0	15.0%	5.0%	$1.0 \times 15.0\% + 0.0 \times 5.0\%$	$= 15\%$
800	200	17.5%	5.0%	$0.8 \times 17.5\% + 0.2 \times 5.0\%$	$= 15\%$
500	500	25.0%	5.0%	$0.5 \times 25.0\% + 0.5 \times 5.0\%$	$= 15\%$
100	900	75.0%	8.3%*	$0.1 \times 75.0\% + 0.9 \times 8.3\%$	$= 15\%$

(b)

Depreciation tax shield and interest tax shield

In chapter 8 you have come across the notion of a depreciation tax shield. Similarly, there are interest tax shields since corporate taxes are paid after interest is deducted from EBIT. This leads to interest rate tax shields (See Chapter 12.6), a notion that we can appreciate without reading Chapter 15 of the text book.

- Free Cash Flow

$$\text{Free Cash Flow} = \overbrace{(\text{Revenues} - \text{Costs} - \text{Depreciation}) \times (1 - \tau_c)}^{\text{Unlevered Net Income}} + \text{Depreciation} - \text{CapEx} - \Delta NWC \quad (\text{Eq. 8.5})$$

$$\text{Free Cash Flow} = (\text{Revenues} - \text{Costs}) \times (1 - \tau_c) - \text{CapEx} - \Delta NWC + \tau_c \times \text{Depreciation} \quad (\text{Eq. 8.6})$$

– The term $\tau_c \times \text{Depreciation}$ is called the depreciation tax shield

Interest tax shield

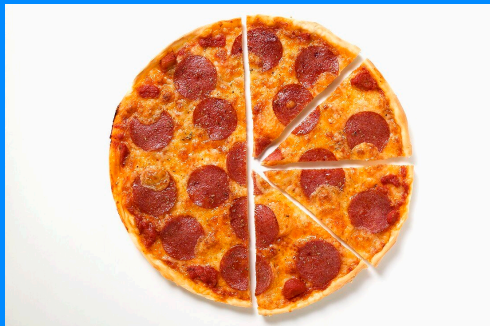


TABLE 15.1

**Macy's Income with and without Leverage,
Fiscal Year 2014 (\$ million)**

	With Leverage	Without Leverage
EBIT	\$2800	\$2800
Interest expense	−400	0
Income before tax	2400	2800
Taxes (35%)	−840	−980
Net income	\$1560	\$1820

	With Leverage	Without Leverage
Interest paid to debt holders	400	0
Income available to equity holders	1560	1820
Total available to all investors	\$1960	\$1820

Gain in net income is equal to reduction in taxes with leverage: \$980m-\$840m = \$140m

Macy's does not owe taxes on \$400m of earnings which are paid as interest, so this \$400 is *shielded* from corporate tax providing the tax savings of $35\% \times \$400\text{m} = \140m

(Annual) interest tax shield = $\tau_c \times \text{Interest Payments}$

The Interest Tax Shield with Permanent Debt

- Typically, the level of future interest payments is uncertain due to changes in the marginal tax rate, the amount of debt outstanding, the interest rate on that debt, and the risk of the firm.
 - For simplicity, we will consider the special case in which the above variables are kept constant
 - Suppose a firm borrows debt D and keeps the debt permanently. If the firm's marginal tax rate is τ_c , and if the debt is riskless with a risk-free interest rate r_f , then the interest tax shield each year is $\tau_c \times r_f \times D$, and the tax shield can be valued as a perpetuity:

$$\begin{aligned} PV(\text{Interest Tax Shield}) &= \frac{\tau_c \times \text{Interest}}{r_f} = \frac{\tau_c \times (r_f \times D)}{r_f} \\ &= \tau_c \times D \end{aligned}$$

- If the debt is fairly priced, no arbitrage implies that its market value must equal the present value of the future interest payments:

$$\text{Market Value of Debt} = D = PV(\text{Future Interest Payments})$$

- If the firm's marginal tax rate is constant, then

$$\begin{aligned} PV(\text{Interest Tax Shield}) &= PV(\tau_c \times \text{Future Interest Payments}) \\ &= \tau_c \times PV(\text{Future Interest Payments}) \\ &= \tau_c \times D \end{aligned}$$

15.3 Recapitalizing to Capture the Tax Shield

- Assume that Midco Industries wants to boost its stock price.
- The company currently has 20 million shares outstanding with a market price of \$15 per share and no debt.
- Midco has had consistently stable earnings and pays a 21% tax rate.
- Management plans to borrow \$100 million on a permanent basis, and they will use the borrowed funds to repurchase outstanding shares.

15.3 Recapitalizing to Capture the Tax Shield

- Without leverage
 - $V_U = (20 \text{ million shares}) \times (\$15 \text{ per share}) = \300 million
- If Midco borrows \$100 million using permanent debt, the present value of the firm's future tax savings is

$$PV(\text{InterestTaxShield}) = \tau_c D = 21\% \times \$100\text{million} = \$21\text{million}$$

Thus the total value of the levered firm will be

$$V^L = V^U + \tau_c D = \$300\text{million} + \$21\text{million} = \$321\text{million}$$

Because the value of the debt is \$100 million, the value of the equity is

$$E = V^L - D = \$321\text{million} - \$100\text{million} = \$221\text{million}$$

- Although the value of the shares outstanding drops to \$221 million, shareholders will also receive the \$100 million that Midco will pay out through the share repurchase.
- In total, they will receive the full \$321 million, a gain of \$21 million over the value of their shares without leverage.

12.6 Financing and the Weighted Average Cost of Capital

- How might the project's cost of capital change if the firm uses leverage to finance the project?
- Perfect Capital Markets
 - In perfect capital markets, choice of financing does not affect cost of capital or project NPV
- Taxes – A Big Imperfection
 - When interest payments on debt are tax deductible, the net cost to the firm is given by

$$\text{Effective after-tax interest rate} = r(1 - \tau_c)$$

- The Weighted Average Cost of Capital (WACC)

$$r_{wacc} = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - \tau_c)$$

- Given a target leverage ratio,

$$r_{wacc} = r_U - \frac{D}{E + D} \tau_c r_D$$

The Weighted Average Cost of Capital

Estimating the WACC

- How does r_{wacc} compare with r_U ?
 - Unlevered cost of capital (or pretax WACC)
 - Expected return investors will earn by holding the firm's assets
 - In a world with taxes, it can be used to evaluate an all-equity project with the same risk as the firm
 - In a world with taxes, WACC is less than the expected return of the firm's assets.
 - With taxes, WACC can be used to evaluate a project with the same risk and the same financing as the firm.

Problem

Dunlap Corp. has a market capitalization of \$100 million, and \$25 million in outstanding debt.

Dunlap's equity cost of capital is 10%, and its debt cost of capital is 6%. What is Dunlap's unlevered cost of capital?

If its corporate tax rate is 25%, what is Dunlap's weighted average cost of capital?

Textbook Example 12.9

Solution

Dunlap's unlevered cost of capital, or pretax WACC, is given by

$$r_U =$$

Thus, we would use a cost of capital of% to evaluate all-equity financed project with the same risk as Dunlap's assets.

Dunlap's weighted average cost of capital, or WACC, can be calculated using either Eq.12.12 or 12.13:

$$r_{wacc} = \dots$$

We can use the WACC ofto evaluate with the same risk and the same mix of debt and equity financing as Dunlap's assets. It is a lower rate than the unlevered cost of capital to reflect the tax deductibility of interest expenses.

Main Takeaways Week 5

- How to estimate the cost of debt
- Calculate the weighted average cost of capital.
- Describe the capital structure that the firm should choose.
- List the three conditions that make capital markets perfect.
- Discuss the implications of MM Proposition I, and the roles of homemade leverage and the Law of One Price in the development of the proposition.
- Calculate the cost of capital for levered equity according to MM Proposition II.

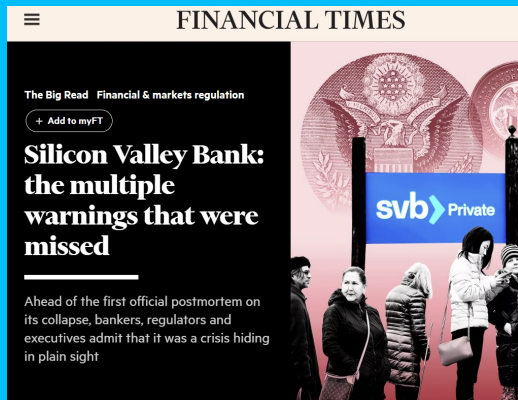
Appendix:

Understand differences in the balance sheets and risks faced by insurers and banks

Revision of
material covered
in weeks 1-5

Any questions?

Appendix A: Balance Sheets of Banks and Insurance Companies



- We have so far studied balance sheets of typical firms
- Since in October-December you will study the behavior of banks and other institutional investors it is important to introduce their balance sheets and understand how these affect their decision making
- In the next slides we will review the balance sheet of a bank and insurance company
- The material in these slides will however not appear in the tutorials or final exam
- Background readings:
 - <https://www.insuranceeurope.eu/publications/488/why-insurers-differ-from-banks/>
 - <https://www.ft.com/content/e128f83e-0707-476b-a8f0-255f2a9bd881>
 - https://www.bankingsupervision.europa.eu/about/ssmexplained/html/hold_capital.en.html

Key Distinction: Life and Non-Life Insurance

Non-life insurers must meet liabilities (accident claims) of a **much shorter duration** due to the typical three to five-year underwriting cycle.

The business cycle tends to drive a company's need for liquidity.

Interest rate risk is less of a consideration for a non-life insurer than a life insurer.

Figure 2: European health, P&C and accident (premiums and claims paid in 2013)



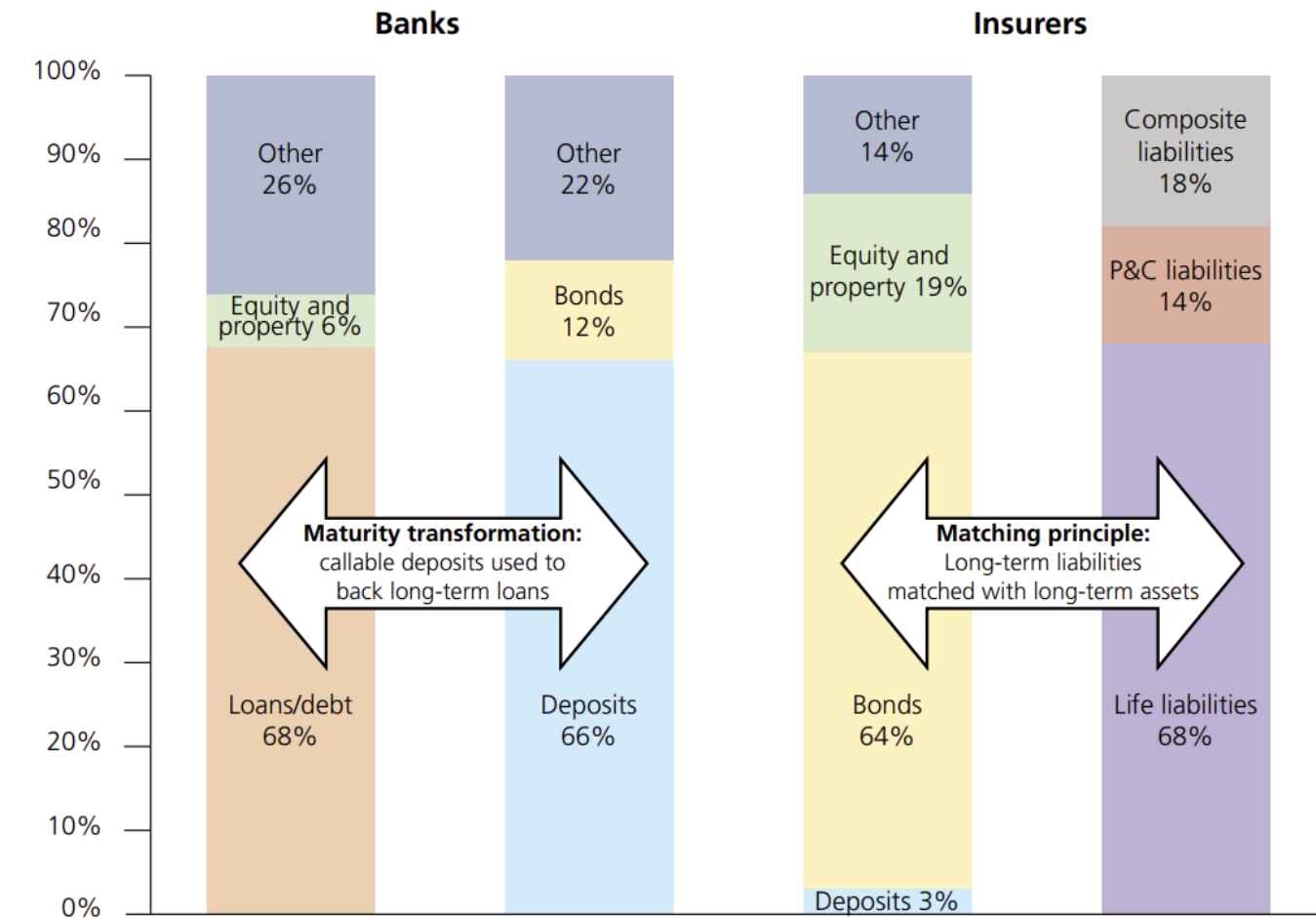
Figure 3: European life insurance premiums and benefits paid in 2013



Source: <https://www.insuranceeurope.eu/publications/488/why-insurers-differ-from-banks/>

Figure 7: How insurer and bank balance sheets differ

Source:
<https://www.insuranceeurope.eu/publications/488/why-insurers-differ-from-banks/>



Sources: ECB; Bank of England; Oliver Wyman analysis

Balance Sheet Differences Between Banks and Insurance Companies

In contrast with bank deposits, the liabilities for insurance of general protection, property, casualty, and health are not callable at will.

1. The balance sheet structures of banks and insurance companies show remarkable differences.
2. The balance sheet of insurers is economically stable, given that fairly long-term policyholder liabilities are matched with assets of corresponding duration.
3. The risks faced by an insurer depend on both assets and liabilities as well as on the way these two interact.
4. In the case of banks, assets and liabilities are in principle not strictly “linked” and banks engage in **maturity transformation**, so that the average duration of most banks’ assets is longer than the average duration of their liabilities.
5. Main Risks Facing Insurers and Banks:

Insurers	Banks
Insurance Risk/Underwriting Risk	Credit Risk
Investment / Market Risk	Liquidity Risk
Mismatch Risk	Market Risk
	Compliance Risk

Appendix: 12.5 A Project's Cost of Capital

Textbook Example 12.5

Unlevering the Cost of Capital Problem

Your firm is considering expanding its household products division.

You identify Procter & Gamble (PG) as a firm with comparable investments.

Suppose PG's equity has a market capitalization of \$144 billion and a beta of 0.55. PG also has \$37 billion of AA-rated debt outstanding, with an average yield of 3.1%.

Estimate the cost of capital of your firm's investment given a risk-free rate of 3% and a market risk-premium of 5%.

12.5 A Project's Cost of Capital

Textbook Example 12.5

Solution

Investing in the division is like investing in PG's assets by holding its debt and equity.

First, we estimate PG's equity cost of capital based on PG's unlevered cost of capital:

$R_E =$

We estimate PG's equity cost of capital using the CAPM as $R_E =$

Because PG's debt is highly rated, we approximate its debt cost of capital using the debt yield of 3.1%.

Thus, PG's unlevered cost of capital is

$$r_u =$$

Alternative, we can estimate PG's unlevered beta. Given its high rating, if we assume PG's debt is zero we have

$$\beta_U =$$

Taking this result as an estimate of the beta of our project, we can compute our project's cost of capital from the CAPM as

$$r_U = \dots$$

Example

Problem

Suppose the entrepreneur of Section 14.1 borrows only \$200 when financing the project. According to MM Proposition II, **what will be the firm's equity cost of capital?**

Solution

Because the firm's assets have a market value of _____, by MM Proposition I the equity will have a market value of _____.

Then, using Eq. 14.5,

$$r_E =$$